

TECHNICAL APTITUDE (BASIC PROGRAMMING AND AIML CONCEPTS)

Full Question Bank
Questions, Solutions, and Explanations

=====

SECTION: 01 Basic

=====

--- README.md ---

100 Technical Aptitude Practice Questions (Basic Level) with Solutions

This file contains 100 basic-level practice questions for Technical Aptitude, designed to develop foundational technical knowledge and problem-solving skills for fresher job aptitude tests, especially for AI/ML roles. Each question includes a description of the problem (e.g., programming fundamentals, data structures, algorithms, databases, networks, operating systems, AI/ML basics) and a detailed solution explaining the reasoning process. Topics cover basic syntax, control structures, arrays, simple algorithms, SQL queries, network protocols, process management, and introductory AI/ML concepts.

Programming Fundamentals (1?20)

1. Question: What is the output of this C code: `int x = 5; printf("%d", x++);`?

Solution:

`x++` is post-increment, so `printf` uses the current value of `x` (5), then increments `x` to 6.

Answer: 5

2. Question: In Python, what is the output of: `print(3 * "hello")`?

Solution:

String repetition in Python multiplies the string: `3 * "hello"` produces `"hellohellohello"`.

Answer: hellohellohello

3. Question: What is the value of `x` after this Java code: `int x = 10; x += 5;`?

Solution:

`x += 5` is equivalent to `x = x + 5`. So, `x = 10 + 5 = 15`.

Answer: 15

4. Question: In C++, what is the output of: `int a = 3, b = 4; cout << a + b;`?

Solution:

`a + b = 3 + 4 = 7`. `cout` prints `7`.

Answer: 7

5. Question: What does this Python code do: `x = [1, 2, 3]; x.append(4)`?

Solution:

`append(4)` adds `4` to the end of the list `x`. New list: `[1, 2, 3, 4]`.

Answer: Adds 4 to the list

6. Question: In C, what is the output of: `int x = 8; printf("%d", x >> 1);`?

Solution:

`x >> 1` performs a right shift. `8` (binary `1000`) shifts to `0100` (4).

Answer: 4

7. Question: In Java, what is the output of: `boolean x = true; System.out.println(!x);`?

Solution:

`!x` negates `true` to `false`.

Answer: false

8. Question: What is the output of this Python code: `x = 10; y = 3; print(x // y)`?

Solution:

`//` is floor division: `10 // 3 = 3` (quotient without remainder).

Answer: 3

9. Question: In C++, what is the output of: `int x = 5; if(x > 3) cout << "Yes"; else cout << "No";`?

Solution:

`x = 5 > 3`, so `if` condition is true. Prints `Yes`.

Answer: Yes

10. Question: In Python, what is the output of: `for i in range(3): print(i)`?

Solution:

`range(3)` generates `0, 1, 2`. Loop prints each value.

Answer:

0

1

2

11. Question: In C, what is the output of: `int x = 3; printf("%d", x & 1);`?

Solution:

`x & 1` performs bitwise AND. `3` (binary `11`) AND `1` (binary `01`) = `01` (1).

Answer: 1

12. Question: In Java, what is the output of: `int x = 7; System.out.println(x % 2);`?

Solution:

`%` is modulus: `7 % 2 = 1` (remainder).

Answer: 1

13. Question: In Python, what is the output of: `x = [1, 2]; x.pop(0); print(x)`?

Solution:

`pop(0)` removes the element at index `0` (1). New list: `[2]`.

Answer: [2]

14. Question: In C++, what is the output of: `int x = 4; cout << ++x;`?

Solution:

`++x` is pre-increment, so `x` becomes `5`, and `5` is printed.

Answer: 5

15. Question: In C, what is the output of: `int x = 5; printf("%d", x | 2);`?

Solution:

`x | 2` performs bitwise OR. `5` (binary `101`) OR `2` (binary `010`) = `111` (7).

Answer: 7

16. Question: In Python, what is the output of: `x = {1, 2, 2}; print(len(x))`?

Solution:

Sets remove duplicates: `{1, 2, 2}` becomes `{1, 2}`. `len` returns `2`.

Answer: 2

17. Question: In Java, what is the output of: `String s = "Hello"; System.out.println(s.length());`?

Solution:

`length()` returns the number of characters in `"Hello"`, which is `5`.

Answer: 5

18. Question: In C++, what is the output of: `int x = 10; while(x > 8) { cout << x << " "; x--; }`?

Solution:

Loop runs while `x > 8`: prints `10`, then `9`.

Answer: 10 9

19. Question: In Python, what is the output of: `x = 5; print(x ** 2)`?

Solution:

`**` is exponentiation: `5 ** 2 = 25`.

Answer: 25

20. Question: In C, what is the output of: `int x = 6; printf("%d", x ^ 3);`?

Solution:

`x ^ 3` performs bitwise XOR. `6` (binary `110`) XOR `3` (binary `011`) = `101` (5).

Answer: 5

Data Structures (21?40)

21. Question: What is the time complexity of accessing an element in an array?

Solution:

Arrays allow direct index-based access, which is constant time.

Answer: O(1)

22. Question: In a singly linked list, what is the time complexity to insert a node at the beginning?

Solution:

Inserting at the head updates the head pointer, which is constant time.

Answer: $O(1)$

23. Question: What is the output of this Python code for a stack: `s = []; s.append(1); s.append(2); s.pop(); print(s)`?

Solution:

Stack (LIFO): `append` adds `1`, `2`. `pop` removes `2`. Stack: `[1]`.

Answer: `[1]`

24. Question: In a queue, what is the order of elements after: `enqueue(1)`, `enqueue(2)`, `dequeue()`, `enqueue(3)`?

Solution:

Queue (FIFO): `Enqueue 1`, `2`. `Dequeue` removes `1`. `Enqueue 3`. Queue: `2, 3`.

Answer: 2, 3

25. Question: What is the time complexity of searching an element in an unsorted array of size n ?

Solution:

Linear search checks each element, worst case n checks.

Answer: $O(n)$

26. Question: In a binary tree, what is the maximum number of nodes at level k (root at level 0)?

Solution:

At level k , maximum nodes = 2^k .

Answer: 2^k

27. Question: What is the output of this Python code for a list: `x = [1, 2, 3]; x.insert(1, 4); print(x)`?

Solution:

`insert(1, 4)` adds `4` at index `1`. New list: `[1, 4, 2, 3]`.

Answer: `[1, 4, 2, 3]`

28. Question: What is the time complexity of inserting an element at the end of a dynamic array (amortized)?

Solution:

Appending to a dynamic array is $O(1)$ amortized, as resizing is rare.

Answer: $O(1)$

29. Question: In a circular queue with size 5, after `enqueue(1, 2, 3)`, `dequeue()`, `enqueue(4)`, what is the front element?

Solution:

`Enqueue 1, 2, 3`. `Dequeue` removes `1`. `Enqueue 4`. Front: `2`.

Answer: 2

30. Question: What is the height of a binary tree with 7 nodes in a balanced configuration?

Solution:

Balanced binary tree: height = $\lceil \log_2(7) \rceil = 3$ (levels 0, 1, 2).

Answer: 2

31. Question: In Python, what is the output of: `d = {1: 'a', 2: 'b'}; print(d[1])`?`

Solution:

Dictionary access by key: `d[1]` returns `'a'`.

Answer: a

32. Question: What is the time complexity of deleting the last node in a singly linked list with n nodes?

Solution:

Traverse to the second-to-last node to update the pointer: $O(n)$.

Answer: $O(n)$

33. Question: In a stack, after push(1), push(2), pop(), push(3), what is the top element?

Solution:

Stack (LIFO): Push `'1'`, `'2'`. Pop removes `'2'`. Push `'3'`. Top: `'3'`.

Answer: 3

34. Question: What is the time complexity of searching in a binary search tree with n nodes (balanced)?

Solution:

Balanced BST search follows height, which is $O(\log n)$.

Answer: $O(\log n)$

35. Question: In Python, what is the output of: `s = {1, 2, 3}; s.add(4); print(s)`?`

Solution:

`add(4)` adds `'4'` to the set. New set: `{1, 2, 3, 4}`.

Answer: {1, 2, 3, 4}

36. Question: What is the space complexity of storing a binary tree with n nodes?

Solution:

Each node stores data and two pointers, total $O(n)$ space.

Answer: $O(n)$

37. Question: In a queue, after enqueue(1, 2), dequeue(), enqueue(3, 4), dequeue(), what is the front element?

Solution:

Enqueue `'1'`, `'2'`. Dequeue removes `'1'`. Enqueue `'3'`, `'4'`. Dequeue removes `'2'`. Front: `'3'`.

Answer: 3

38. Question: What is the time complexity of inserting a key into a hash table (average case)?

Solution:

Average case with good hash function: $O(1)$ for insertion.

Answer: $O(1)$

39. Question: In Python, what is the output of: `x = [1, 2, 3]; x.remove(2); print(x)`?

Solution:

`remove(2)` deletes the first occurrence of `2`. New list: `[1, 3]`.

Answer: `[1, 3]`

40. Question: What is the maximum number of nodes in a binary tree of height h ?

Solution:

Maximum nodes = $2^{(h+1)} - 1$ (sum of nodes from level 0 to h).

Answer: $2^{(h+1)} - 1$

Algorithms (41?60)

41. Question: What is the time complexity of bubble sort for n elements?

Solution:

Bubble sort compares and swaps in nested loops: $O(n^2)$.

Answer: $O(n^2)$

42. Question: In Python, what is the output of this linear search: `x = [3, 1, 4]; print(x.index(4))`?

Solution:

`index(4)` returns the first index of `4`, which is `2`.

Answer: 2

43. Question: What is the time complexity of binary search on a sorted array of n elements?

Solution:

Binary search halves the search space each step: $O(\log n)$.

Answer: $O(\log n)$

44. Question: What is the output of this C code for factorial: `int fact(int n) { if(n == 0) return 1; return n * fact(n-1); } printf("%d", fact(3));`?

Solution:

`fact(3)` = $3 \times \text{fact}(2)$ = $3 \times 2 \times \text{fact}(1)$ = $3 \times 2 \times 1 = 6$.

Answer: 6

45. Question: What is the space complexity of a recursive factorial algorithm?

Solution:

Recursive factorial uses $O(n)$ stack space for n calls.

Answer: $O(n)$

46. Question: In Python, what is the output of: `x = [5, 2, 8]; print(sorted(x))`?

Solution:

`sorted(x)` returns a new sorted list: `[2, 5, 8]`.

Answer: `[2, 5, 8]`

47. Question: What is the time complexity of finding the maximum element in an unsorted array of n elements?

Solution:

Scan all elements: $O(n)$.

Answer: $O(n)$

48. Question: In C, what is the output of this sum code: ``int sum(int n) { if(n == 0) return 0; return n + sum(n-1); } printf("%d", sum(4));``?

Solution:

``sum(4)`` = `4 + `sum(3)`` = `4 + 3 + `sum(2)`` = `4 + 3 + 2 + `sum(1)`` = `4 + 3 + 2 + 1` = 10.

Answer: 10

49. Question: What is the time complexity of selection sort for n elements?

Solution:

Selection sort finds the minimum in each pass: $O(n^2)$.

Answer: $O(n^2)$

50. Question: In Python, what is the output of: ``x = [1, 2, 3]; print(x[::-1])``?

Solution:

Slicing ``[::-1]`` reverses the list: ``[3, 2, 1]``.

Answer: [3, 2, 1]

51. Question: What is the time complexity of checking if an element exists in a sorted array using binary search?

Solution:

Binary search: $O(\log n)$.

Answer: $O(\log n)$

52. Question: In C++, what is the output of: ``int x = 5; for(int i = 1; i <= 3; i++) x += i; cout << x;``?

Solution:

Loop: ``x = 5 + 1 + 2 + 3 = 11``.

Answer: 11

53. Question: What is the space complexity of bubble sort?

Solution:

Bubble sort uses a constant amount of extra space: $O(1)$.

Answer: $O(1)$

54. Question: In Python, what is the output of: ``x = [1, 2, 3]; print(sum(x))``?

Solution:

``sum(x)`` adds elements: ``1 + 2 + 3 = 6``.

Answer: 6

55. Question: What is the time complexity of inserting an element into a sorted array of n elements?

Solution:

Shift elements to make space: $O(n)$.

Answer: $O(n)$

56. Question: In C, what is the output of: `int a = 3, b = 5; if(a > b) printf("%d", a); else printf("%d", b);`?

Solution:

`3 > 5` is false, so print `b = 5`.

Answer: 5

57. Question: What is the time complexity of merging two sorted arrays of sizes m and n ?

Solution:

Merge process compares and copies: $O(m + n)$.

Answer: $O(m + n)$

58. Question: In Python, what is the output of: `x = "abc"; print(x.upper())`?

Solution:

`upper()` converts to uppercase: `"ABC"`.

Answer: ABC

59. Question: What is the time complexity of reversing an array of n elements?

Solution:

Swap elements from ends to middle: $O(n)$.

Answer: $O(n)$

60. Question: In C++, what is the output of: `int x = 4; x *= 3; cout << x;`?

Solution:

`x *= 3` means `x = x * 3 = 4 * 3 = 12`.

Answer: 12

Database Basics (61-70)

61. Question: What does the SQL query `SELECT * FROM table_name;` do?

Solution:

Retrieves all columns and rows from `table_name`.

Answer: Fetches all data

62. Question: What is the output of: `SELECT name FROM employees WHERE salary > 50000;`?

Solution:

Returns the `name` column for rows where `salary` exceeds 50000.

Answer: Names of employees with salary > 50000

63. Question: What does the SQL keyword `PRIMARY KEY` ensure?

Solution:

Ensures each value in the column is unique and not null.

Answer: Uniqueness and non-null

64. Question: What is the result of: `SELECT COUNT(\*) FROM students WHERE age = 20;`?

Solution:

Counts the number of rows where `age` is 20.

Answer: Number of students aged 20

65. Question: What does `DELETE FROM table\_name WHERE id = 1;` do?

Solution:

Removes the row where `id` equals 1.

Answer: Deletes row with id = 1

66. Question: What is the purpose of the SQL `JOIN` clause?

Solution:

Combines rows from two or more tables based on a related column.

Answer: Merges tables

67. Question: What does `INSERT INTO users (name, age) VALUES ('Alice', 25);` do?

Solution:

Adds a new row to `users` with `name = 'Alice'` and `age = 25`.

Answer: Inserts a new user

68. Question: What is the output of: `SELECT MAX(salary) FROM employees;`?

Solution:

Returns the highest value in the `salary` column.

Answer: Maximum salary

69. Question: What does the SQL `WHERE` clause do?

Solution:

Filters rows based on a specified condition.

Answer: Filters data

70. Question: What is the result of: `UPDATE products SET price = 100 WHERE id = 5;`?

Solution:

Sets `price` to 100 for the row where `id` is 5.

Answer: Updates price for id = 5

Computer Networks (71?80)

71. Question: What is the purpose of the IP address?

Solution:

Uniquely identifies a device on a network.

Answer: Device identification

72. Question: What is the port number used by HTTP?

Solution:

HTTP uses port 80 for communication.

Answer: 80

73. Question: What does the acronym TCP stand for?

Solution:

Transmission Control Protocol, ensures reliable data transfer.

Answer: Transmission Control Protocol

74. Question: What is the role of DNS in networking?

Solution:

Translates domain names (e.g., google.com) to IP addresses.

Answer: Domain name resolution

75. Question: What is the difference between IPv4 and IPv6?

Solution:

IPv4 uses 32-bit addresses (e.g., 192.168.0.1); IPv6 uses 128-bit addresses for more capacity.

Answer: Address length and capacity

76. Question: What does the term "bandwidth" refer to?

Solution:

Maximum data transfer rate of a network connection.

Answer: Data transfer rate

77. Question: What is the purpose of a router in a network?

Solution:

Forwards data packets between networks.

Answer: Routes data

78. Question: What is the default subnet mask for a Class C IP address?

Solution:

Class C uses 24 bits for the network: 255.255.255.0.

Answer: 255.255.255.0

79. Question: What is the function of the ARP protocol?

Solution:

Maps IP addresses to MAC addresses.

Answer: Address resolution

80. Question: What does UDP stand for, and how does it differ from TCP?

Solution:

User Datagram Protocol; UDP is faster but less reliable than TCP (no error checking).

Answer: User Datagram Protocol, less reliable

Operating Systems (81?90)

81. Question: What is the role of an operating system?

Solution:

Manages hardware, software, and provides user interface.

Answer: Resource management

82. Question: What is a process in an operating system?

Solution:

A program in execution, with its own memory and resources.

Answer: Executing program

83. Question: What is the purpose of a scheduler in an OS?

Solution:

Allocates CPU time to processes for efficient execution.

Answer: CPU allocation

84. Question: What is a deadlock in an operating system?

Solution:

A situation where processes hold resources and wait for others, causing a standstill.

Answer: Resource conflict standstill

85. Question: What does the term "virtual memory" refer to?

Solution:

Uses disk space to extend RAM, allowing larger programs to run.

Answer: Extended memory

86. Question: What is the function of a file system in an OS?

Solution:

Organizes and manages data storage on disks.

Answer: Data organization

87. Question: What is a thread in an operating system?

Solution:

A lightweight process sharing the same memory space as its parent process.

Answer: Lightweight process

88. Question: What is the purpose of an interrupt in an OS?

Solution:

Signals the CPU to handle an urgent event (e.g., hardware input).

Answer: Event handling

89. Question: What is the difference between a kernel and a shell?

Solution:

Kernel manages system resources; shell is the user interface to interact with the OS.

Answer: Kernel: core, Shell: interface

90. Question: What is the role of a page table in virtual memory?

Solution:

Maps virtual addresses to physical addresses.

Answer: Address mapping

AI/ML Foundations (91?100)

91. Question: What is supervised learning in machine learning?

Solution:

Model trains on labeled data (input-output pairs) to predict outputs.

Answer: Learning with labeled data

92. Question: What is the purpose of a loss function in ML?

Solution:

Measures the error between predicted and actual outputs to guide optimization.

Answer: Error measurement

93. Question: What is overfitting in machine learning?

Solution:

Model learns training data too well, including noise, and performs poorly on new data.

Answer: Excessive training fit

94. Question: What is the role of a feature in ML?

Solution:

An input variable used to make predictions.

Answer: Input variable

95. Question: What is the purpose of splitting data into training and testing sets?

Solution:

Training set trains the model; testing set evaluates its performance on unseen data.

Answer: Model training and evaluation

96. Question: What is a neuron in a neural network?

Solution:

A computational unit that processes inputs, applies weights, and produces an output.

Answer: Computational unit

97. Question: What is the activation function in a neural network?

Solution:

Introduces non-linearity to decide whether a neuron should activate.

Answer: Non-linearity function

98. Question: What is gradient descent in ML?

Solution:

Optimization algorithm to minimize the loss function by adjusting model parameters.

Answer: Parameter optimization

99. Question: What is unsupervised learning in ML?

Solution:

Model finds patterns in unlabeled data without predefined outputs.

Answer: Learning without labels

100. Question: What is the purpose of normalization in ML?

Solution:

Scales features to a standard range to improve model performance.

Answer: Feature scaling

=====
SECTION: 02 Intermediate
=====

--- README.md ---

100 Technical Aptitude Practice Questions (Intermediate Level) with Solutions

This file contains 100 intermediate-level practice questions for Technical Aptitude, designed to enhance technical knowledge and problem-solving skills for fresher job aptitude tests, especially for AI/ML roles. Each question includes a detailed description of the problem (e.g., programming, data structures, algorithms, databases, networks, operating systems, AI/ML) and a comprehensive solution explaining the reasoning process. Topics cover moderately complex programming logic, linked lists, trees, sorting algorithms, SQL joins, network layers, process synchronization, and AI/ML model evaluation.

Programming Fundamentals (1?20)

1. Question: What is the output of this C code: ``int x = 5; printf("%d %d", x++, ++x);``?

Solution:

In ``printf``, arguments are evaluated right-to-left. ``++x`` increments ``x`` to 6, then ``x++`` uses 6 and increments ``x`` to 7. Output: ``6 6``.

Answer: 6 6

2. Question: In Python, what is the output of: ``x = [1, 2, 3]; y = x; y[0] = 4; print(x)``?

Solution:

``y = x`` creates a reference to the same list. Modifying ``y[0]`` changes ``x``. Output: ``[4, 2, 3]``.

Answer: [4, 2, 3]

3. Question: In Java, what is the output of: ``String s = "hello"; s = s.concat(" world");``

`System.out.println(s);`?`

Solution:

``concat`` creates a new string ``"hello world"``. Output: ``hello world``.

Answer: hello world

4. Question: In C++, what is the output of: ``int x = 10; cout << (x > 5 ? x : 5);`?`

Solution:

Ternary operator: ``x > 5`` is true, so prints ``x = 10``.

Answer: 10

5. Question: In Python, what is the output of: ``def f(x): x += 1; x = 5; f(x); print(x)`?`

Solution:

Python passes integers by value. ``x`` inside ``f`` is local. Output: ``5``.

Answer: 5

6. Question: In C, what is the output of: ``int x = 10; printf("%d", x << 2);`?`

Solution:

``x << 2`` left-shifts ``10`` (binary ``1010``) to ``101000`` (40).

Answer: 40

7. Question: In Java, what is the output of: ``int[] arr = {1, 2, 3}; System.out.println(arr[arr.length - 1]);`?`

Solution:

``arr.length - 1 = 2``. ``arr[2] = 3``.

Answer: 3

8. Question: In Python, what is the output of: ``x = {1: 'a', 2: 'b'}; x.update({2: 'c'}); print(x)`?`

Solution:

``update`` modifies key ``2`` to ``c``. Output: ``{1: 'a', 2: 'c'}``.

Answer: {1: 'a', 2: 'c'}

9. Question: In C++, what is the output of: ``int x = 5; for(int i = 0; i < 3; i++) x *= 2; cout << x;`?`

Solution:

Loop: ``x = 5 * 2 * 2 * 2 = 40``.

Answer: 40

10. Question: In C, what is the output of: ``int a = 7, b = 3; printf("%d", a ^ b);`?`

Solution:

``a ^ b``: ``7`` (binary ``111``) XOR ``3`` (binary ``011``) = ``100`` (4).

Answer: 4

11. Question: In Python, what is the output of: ``x = [1, 2, 3]; print(x[1:3])`?`

Solution:

Slicing ``[1:3]`` includes indices 1 to 2: ``[2, 3]``.

Answer: [2, 3]

12. Question: In Java, what is the output of: `int x = 15; System.out.println(x & (x - 1));`?

Solution:

`x & (x - 1)` clears the least significant set bit. `15` (binary `1111`) & `14` (binary `1110`) = `1110` (14).

Answer: 14

13. Question: In C++, what is the output of: `string s = "abc"; reverse(s.begin(), s.end()); cout << s;`?

Solution:

`reverse` reverses the string: `"abc"` becomes `"cba"`.

Answer: cba

14. Question: In Python, what is the output of: `x = lambda a: a * 2; print(x(3))`?

Solution:

Lambda function doubles input: `3 * 2 = 6`.

Answer: 6

15. Question: In C, what is the output of: `int x = 5; printf("%d", ~x);`?

Solution:

`~x` inverts bits of `5` (binary `00000101`) to `11111010` (two's complement, -6).

Answer: -6

16. Question: In Java, what is the output of: `int x = 8; System.out.println(x >> 1);`?

Solution:

`x >> 1` right-shifts `8` (binary `1000`) to `0100` (4).

Answer: 4

17. Question: In Python, what is the output of: `x = [1, 2]; y = x.copy(); y[0] = 3; print(x)`?

Solution:

`copy()` creates a shallow copy. Modifying `y` doesn't affect `x`. Output: `[1, 2]`.

Answer: [1, 2]

18. Question: In C++, what is the output of: `int x = 3; cout << (x << 1) + (x >> 1);`?

Solution:

`x << 1 = 3 * 2 = 6`. `x >> 1 = 3 / 2 = 1`. `6 + 1 = 7`.

Answer: 7

19. Question: In C, what is the output of: `int arr[3] = {1, 2, 3}; printf("%d", *(arr + 1));`?

Solution:

`*(arr + 1)` accesses `arr[1] = 2`.

Answer: 2

20. Question: In Python, what is the output of: `x = [1, 2, 3]; print(list(map(lambda x: x2, x)))`?

Solution:

`map` applies square function: `[12, 22, 32] = [1, 4, 9]`.

Answer: [1, 4, 9]

Data Structures (21?40)

21. Question: What is the time complexity of inserting a node at the end of a singly linked list with n nodes?

Solution:

Traverse to the end to update the pointer: $O(n)$.

Answer: $O(n)$

22. Question: In Python, what is the output of this stack operation: `s = []; s.append(1); s.append(2); s.append(3); s.pop(); s.pop(); print(s)`?

Solution:

Stack (LIFO): Push `1, 2, 3`. Pop removes `3`, then `2`. Stack: `[1]`.

Answer: [1]

23. Question: In a doubly linked list, what is the time complexity to delete a node given its pointer?

Solution:

Update prev/next pointers of adjacent nodes: $O(1)$.

Answer: $O(1)$

24. Question: In a queue, after `enqueue(1, 2, 3)`, `dequeue()`, `enqueue(4)`, `dequeue()`, what is the front element?

Solution:

Queue (FIFO): Enqueue `1, 2, 3`. Dequeue removes `1`. Enqueue `4`. Dequeue removes `2`. Front: `3`.

Answer: 3

25. Question: What is the time complexity of searching in an unbalanced binary search tree with n nodes?

Solution:

Worst case (skewed tree) resembles a linked list: $O(n)$.

Answer: $O(n)$

26. Question: In Python, what is the output of: `from collections import deque; q = deque([1, 2]); q.append(3); q.popleft(); print(q)`?

Solution:

Deque: Append `3`, popleft removes `1`. Deque: `[2, 3]`.

Answer: deque([2, 3])

27. Question: What is the time complexity of inserting an element in a min-heap with n elements?

Solution:

Insert at bottom, bubble up to restore heap property: $O(\log n)$.

Answer: $O(\log n)$

28. Question: In a binary tree, what is the time complexity of finding the height?

Solution:

Recursively visit all nodes: $O(n)$.

Answer: $O(n)$

29. Question: In Python, what is the output of: `d = {'a': 1, 'b': 2}; d.pop('a'); print(d)`?

Solution:

`pop('a')` removes key `'a'`. Output: `{'b': 2}`.

Answer: `{'b': 2}`

30. Question: What is the time complexity of deleting the root of a max-heap with n elements?

Solution:

Replace root with last element, bubble down: $O(\log n)$.

Answer: $O(\log n)$

31. Question: In a circular linked list, what is the time complexity to insert a node after a given node?

Solution:

Update pointers of given and next node: $O(1)$.

Answer: $O(1)$

32. Question: In Python, what is the output of: `s = {1, 2, 3}; s.discard(2); print(s)`?

Solution:

`discard(2)` removes `2`. Output: `{1, 3}`.

Answer: `{1, 3}`

33. Question: In a binary search tree, what is the time complexity to find the minimum element?

Solution:

Traverse to the leftmost node: $O(h)$, where h is height ($O(\log n)$ if balanced).

Answer: $O(\log n)$ (balanced)

34. Question: What is the space complexity of a depth-first search (DFS) on a binary tree with n nodes?

Solution:

Recursion stack uses $O(h)$, where h is height ($O(\log n)$ if balanced).

Answer: $O(\log n)$ (balanced)

35. Question: In Python, what is the output of: `x = [[1, 2], [3, 4]]; x[0].append(5); print(x)`?

Solution:

Append `5` to `x[0]`. Output: `[[1, 2, 5], [3, 4]]`.

Answer: `[[1, 2, 5], [3, 4]]`

36. Question: What is the time complexity of level-order traversal (BFS) of a binary tree with n nodes?

Solution:

Visit all nodes using a queue: $O(n)$.

Answer: $O(n)$

37. Question: In a hash table with chaining, what is the average time complexity for search with n elements and m buckets?

Solution:

Average chain length = n/m . Search: $O(n/m)$. If $m \propto n$, $O(1)$.

Answer: $O(1)$ (assuming $m \propto n$)

38. Question: In Python, what is the output of: ``from collections import Counter; c = Counter([1, 2, 2, 3]); print(c[2])``?

Solution:

``Counter`` counts occurrences. ``c[2]`` = 2 (two ``2`s`).

Answer: 2

39. Question: What is the time complexity to check if a binary tree is balanced?

Solution:

Compute height of each subtree, check difference: $O(n)$.

Answer: $O(n)$

40. Question: In a stack, after push(1, 2, 3), pop(), push(4), pop(), what is the top element?

Solution:

Stack: Push ``1, 2, 3``. Pop removes ``3``. Push ``4``. Pop removes ``4``. Top: ``2``.

Answer: 2

Algorithms (41?60)

41. Question: What is the time complexity of merge sort for n elements?

Solution:

Divide and merge: $O(n \log n)$.

Answer: $O(n \log n)$

42. Question: In Python, what is the output of this binary search: ``def bs(arr, t): l, r = 0, len(arr)-1; while l <= r: m = (l+r)//2; if arr[m] == t: return m; elif arr[m] < t: l = m+1; else: r = m-1; return -1; print(bs([1, 2, 3, 4], 3))``?

Solution:

Binary search finds ``3`` at index ``2``.

Answer: 2

43. Question: What is the space complexity of quicksort?

Solution:

Recursion stack: $O(\log n)$ average, $O(n)$ worst case.

Answer: $O(\log n)$ (average)

44. Question: In C, what is the output of this GCD code: ``int gcd(int a, int b) { if(b == 0) return a;`

`return gcd(b, a % b); } printf("%d", gcd(48, 18));`?`

Solution:

Euclidean algorithm: ``gcd(48, 18) = gcd(18, 12) = gcd(12, 6) = gcd(6, 0) = 6``.

Answer: 6

45. Question: What is the time complexity of finding all pairs in an array of n elements that sum to a given value?

Solution:

Using a hash set: $O(n)$. Without hash set: $O(n^2)$.

Answer: $O(n)$ (with hash set)

46. Question: In Python, what is the output of: ``x = [4, 2, 5, 1]; x.sort(); print(x)`?`

Solution:

``sort()`` sorts in-place: ``[1, 2, 4, 5]``.

Answer: [1, 2, 4, 5]

47. Question: What is the time complexity of Dijkstra's algorithm using a priority queue for a graph with V vertices and E edges?

Solution:

Priority queue operations: $O((V + E) \log V)$.

Answer: $O((V + E) \log V)$

48. Question: In C++, what is the output of: ``vector<int> v = {1, 2, 3}; rotate(v.begin(), v.begin()+1, v.end()); cout << v[0];`?`

Solution:

``rotate`` shifts left by 1: ``[2, 3, 1]``. First element: ``2``.

Answer: 2

49. Question: What is the time complexity of insertion sort for n elements?

Solution:

Compare and shift: $O(n^2)$.

Answer: $O(n^2)$

50. Question: In Python, what is the output of: ``x = [1, 2, 3, 4]; print([i for i in x if i % 2 == 0])`?`

Solution:

List comprehension filters even numbers: ``[2, 4]``.

Answer: [2, 4]

51. Question: What is the time complexity of checking if a string is a palindrome?

Solution:

Compare characters from both ends: $O(n)$.

Answer: $O(n)$

52. Question: In C, what is the output of this Fibonacci code: ``int fib(int n) { if(n <= 1) return n; return fib(n-1) + fib(n-2); } printf("%d", fib(5));`?`

Solution:

`fib(5) = fib(4) + fib(3) = (fib(3) + fib(2)) + (fib(2) + fib(1)) = (2 + 1) + (1 + 1) = 5`.

Answer: 5

53. Question: What is the space complexity of merge sort?

Solution:

Extra array for merging: $O(n)$.

Answer: $O(n)$

54. Question: In Python, what is the output of: `x = "hello"; print("".join(reversed(x)))`?

Solution:

`reversed` reverses string: `"olleh"`.

Answer: olleh

55. Question: What is the time complexity of finding the longest common prefix of n strings of length m ?

Solution:

Compare characters across strings: $O(n * m)$.

Answer: $O(n * m)$

56. Question: In C++, what is the output of: `int arr[] = {1, 2, 3}; int *p = arr; cout << *(p + 2);`?

Solution:

`*(p + 2)` accesses `arr[2] = 3`.

Answer: 3

57. Question: What is the time complexity of counting sort for n elements with range k ?

Solution:

Create count array and output: $O(n + k)$.

Answer: $O(n + k)$

58. Question: In Python, what is the output of: `x = [1, 2, 2, 3]; print(list(set(x)))`?

Solution:

`set(x)` removes duplicates, `list` converts back: `[1, 2, 3]`.

Answer: [1, 2, 3]

59. Question: What is the time complexity of detecting a cycle in a linked list?

Solution:

Floyd's cycle-finding algorithm (two pointers): $O(n)$.

Answer: $O(n)$

60. Question: In C, what is the output of: `int arr[] = {3, 1, 4}; qsort(arr, 3, sizeof(int), [](const void* a, const void* b){return *(int*)a - *(int*)b;}); printf("%d", arr[0]);`?

Solution:

`qsort` sorts array: `[1, 3, 4]`. First element: `1`.

Answer: 1

Database Basics (61?70)

61. Question: What is the result of this SQL query: `SELECT name, salary FROM employees WHERE department = 'IT' ORDER BY salary DESC;`?

Solution:

Retrieves `name` and `salary` for `department = 'IT'`, sorted by `salary` in descending order.

Answer: IT employees sorted by salary (highest first)

62. Question: What does `SELECT \* FROM orders INNER JOIN customers ON orders.customer\_id = customers.id;` do?

Solution:

Combines rows from `orders` and `customers` where `customer\_id` matches `id`.

Answer: Joins orders with customer details

63. Question: What is the output of: `SELECT department, COUNT(\*) FROM employees GROUP BY department;`?

Solution:

Counts employees per `department`.

Answer: Department-wise employee counts

64. Question: What does `FOREIGN KEY` constraint ensure in SQL?

Solution:

Ensures values in a column match values in another table's primary key.

Answer: Referential integrity

65. Question: What is the result of: `SELECT product\_name FROM products WHERE price BETWEEN 50 AND 100;`?

Solution:

Retrieves `product\_name` where `price` is 50 to 100 (inclusive).

Answer: Products priced 50?100

66. Question: What does `SELECT DISTINCT city FROM customers;` do?

Solution:

Returns unique `city` values from `customers`.

Answer: Unique cities

67. Question: What is the output of: `SELECT name FROM students WHERE marks > (SELECT AVG(marks) FROM students);`?

Solution:

Retrieves `name` of students with `marks` above the average.

Answer: Students with above-average marks

68. Question: What does `ALTER TABLE employees ADD COLUMN email VARCHAR(100);` do?

Solution:

Adds an `email` column to `employees` table.

Answer: Adds email column

69. Question: What is the result of: `SELECT order\_id, customer\_name FROM orders LEFT JOIN customers ON orders.customer\_id = customers.id;`?

Solution:

Includes all `orders`, with matching `customer\_name` or NULL if no match.

Answer: All orders with customer names (or NULL)

70. Question: What does `CREATE INDEX idx\_name ON employees(name);` do?

Solution:

Creates an index on `name` column to speed up queries.

Answer: Improves query performance

Computer Networks (71?80)

71. Question: What is the purpose of the OSI model?

Solution:

Provides a framework for network communication with seven layers.

Answer: Standardizes network communication

72. Question: Which layer of the OSI model handles routing of packets?

Solution:

Network layer (Layer 3) manages routing.

Answer: Network layer

73. Question: What is the port number used by HTTPS?

Solution:

HTTPS uses port 443 for secure communication.

Answer: 443

74. Question: What is the purpose of NAT (Network Address Translation)?

Solution:

Maps private IP addresses to a public IP for internet access.

Answer: IP address mapping

75. Question: What is the difference between a hub and a switch?

Solution:

Hub broadcasts data to all ports; switch forwards data to specific devices.

Answer: Hub: broadcasts, Switch: targeted

76. Question: What is the role of the MAC address?

Solution:

Uniquely identifies a device at the data link layer.

Answer: Device identification

77. Question: What does the TCP three-way handshake involve?

Solution:

SYN, SYN-ACK, ACK to establish a connection.

Answer: Connection establishment

78. Question: What is the purpose of ICMP?

Solution:

Handles error messages and diagnostics (e.g., ping).

Answer: Error reporting

79. Question: What is the maximum number of hosts in a subnet with mask 255.255.255.240?

Solution:

$2^{(32-24)} = 2^8 = 256$ addresses (254 hosts, 1 network, 1 broadcast).

Answer: 254

80. Question: What is the role of BGP in networking?

Solution:

Border Gateway Protocol routes data between autonomous systems.

Answer: Inter-domain routing

Operating Systems (81-90)

81. Question: What is the difference between preemptive and non-preemptive scheduling?

Solution:

Preemptive: CPU can interrupt a process; Non-preemptive: process runs to completion.

Answer: Preemptive: interruptible, Non-preemptive: uninterruptible

82. Question: What is a semaphore in an operating system?

Solution:

Synchronization tool to control access to shared resources.

Answer: Synchronization mechanism

83. Question: What is the purpose of paging in memory management?

Solution:

Divides memory into fixed-size pages to simplify allocation.

Answer: Memory allocation

84. Question: What is the difference between a process and a thread?

Solution:

Process: independent program with own memory; Thread: lightweight, shares process memory.

Answer: Process: isolated, Thread: shared memory

85. Question: What is the role of the swap space in an OS?

Solution:

Disk space used to store inactive pages when RAM is full.

Answer: Extends virtual memory

86. Question: What is a context switch in an operating system?

Solution:

Saves current process state and loads another to switch execution.

Answer: Process switching

87. Question: What is the purpose of the LRU page replacement algorithm?

Solution:

Replaces the least recently used page to minimize page faults.

Answer: Optimizes page replacement

88. Question: What is a race condition in an OS?

Solution:

Occurs when multiple processes access shared data concurrently, leading to unpredictable results.

Answer: Concurrent access issue

89. Question: What is the role of the file allocation table (FAT) in a file system?

Solution:

Tracks file locations on disk.

Answer: File location tracking

90. Question: What is the purpose of a mutex in an OS?

Solution:

Ensures mutual exclusion to prevent concurrent access to critical sections.

Answer: Mutual exclusion

AI/ML Foundations (91?100)

91. Question: What is the difference between classification and regression in ML?

Solution:

Classification predicts discrete labels; regression predicts continuous values.

Answer: Classification: labels, Regression: values

92. Question: What is the purpose of cross-validation in ML?

Solution:

Splits data into multiple folds to evaluate model performance robustly.

Answer: Robust model evaluation

93. Question: What is the bias-variance tradeoff in ML?

Solution:

High bias: underfitting; high variance: overfitting. Balance minimizes error.

Answer: Balances underfitting and overfitting

94. Question: What is the role of the learning rate in gradient descent?

Solution:

Controls step size for parameter updates during optimization.

Answer: Step size control

95. Question: What is the purpose of regularization in ML?

Solution:

Adds penalty to model complexity to prevent overfitting.

Answer: Prevents overfitting

96. Question: What is the difference between a perceptron and a multi-layer perceptron (MLP)?

Solution:

Perceptron: single-layer; MLP: multiple layers for non-linear problems.

Answer: Perceptron: single, MLP: multiple layers

97. Question: What is the purpose of the softmax function in neural networks?

Solution:

Converts logits to probabilities for multi-class classification.

Answer: Probability conversion

98. Question: What is the role of k in k-nearest neighbors (k-NN)?

Solution:

Number of neighbors considered for prediction.

Answer: Neighbor count

99. Question: What is the difference between bagging and boosting in ML?

Solution:

Bagging: parallel ensembles (e.g., Random Forest); Boosting: sequential, corrects errors (e.g., AdaBoost).

Answer: Bagging: parallel, Boosting: sequential

100. Question: What is the purpose of principal component analysis (PCA) in ML?

Solution:

Reduces dimensionality by projecting data onto principal components.

Answer: Dimensionality reduction

=====

SECTION: 03 Advance

=====

--- README.md ---

100 Technical Aptitude Practice Questions (Advanced Level) with

Solutions

This file contains 100 advanced-level practice questions for Technical Aptitude, designed to rigorously test and enhance technical knowledge and problem-solving skills for fresher job aptitude tests, especially for AI/ML roles. Each question includes a detailed description of the problem (e.g., complex programming, advanced data structures, algorithms, databases, networks, operating systems, AI/ML) and a comprehensive solution explaining the reasoning process. Topics cover dynamic programming, graph algorithms, advanced SQL, network security, concurrency, and AI/ML model optimization.

Programming Fundamentals (1?20)

1. Question: What is the output of this C code: `int x = 5; int *p = &x; *p += 2; printf("%d %d", x, *p);`?

Solution:

`*p` points to `x`. `*p += 2` modifies `x` to `5 + 2 = 7`. Both `x` and `*p` access the same memory. Output: `7 7`.

Answer: 7 7

2. Question: In Python, what is the output of: `x = [[0]*2]*3; x[0][0] = 1; print(x)`?

Solution:

`[[0]*2]*3` creates three references to the same inner list. Modifying `x[0][0]` affects all rows. Output: `[[1, 0], [1, 0], [1, 0]]`.

Answer: `[[1, 0], [1, 0], [1, 0]]`

3. Question: In Java, what is the output of: `String s1 = new String("test"); String s2 = new String("test"); System.out.println(s1 == s2);`?

Solution:

`==` checks reference equality. `new String` creates distinct objects. Output: `false`.

Answer: false

4. Question: In C++, what is the output of: `int x = 10; auto f = [&x]() { x += 5; return x; }; cout << f() << " " << x;`?

Solution:

Lambda captures `x` by reference. `f()` increments `x` to `15`, returns `15`. `x` is also `15`. Output: `15 15`.

Answer: 15 15

5. Question: In Python, what is the output of: `def f(x=[]): x.append(1); return x; print(f(), f())`?

Solution:

Default list `x` is shared across calls. First `f()`: `x = [1]`. Second `f()`: `x = [1, 1]`. Output: `[1, 1]`.

Answer: `[1] [1, 1]`

6. Question: In C, what is the output of: ``int x = 5; void *p = &x; printf("%d", *(int*)p);``?

Solution:

``p`` is a void pointer to ``x``. Cast to ``int*`` and dereference gives ``5``. Output: ``5``.

Answer: 5

7. Question: In Java, what is the output of: ``Integer a = 127; Integer b = 127; System.out.println(a == b);``?

Solution:

Integer caching for ``-128`` to ``127`` reuses objects. ``a`` and ``b`` reference the same object. Output: ``true``.

Answer: true

8. Question: In Python, what is the output of: ``x = {1: [1], 2: [2]}; y = x.copy(); y[1].append(3); print(x[1])``?

Solution:

``copy()`` creates a shallow copy. ``y[1]`` references the same list as ``x[1]``. Appending ``3`` modifies ``x[1]``. Output: ``[1, 3]``.

Answer: [1, 3]

9. Question: In C++, what is the output of: ``int arr[] = {1, 2, 3}; int *p = arr; cout << *(p + 1) * *(p + 2);``?

Solution:

``*(p + 1) = arr[1] = 2``, ``*(p + 2) = arr[2] = 3``. ``2 * 3 = 6``. Output: ``6``.

Answer: 6

10. Question: In C, what is the output of: ``union U { int x; char c; } u; u.x = 258; printf("%d", u.c);``?

Solution:

Union shares memory. ``x = 258`` (binary ``00000001 00000010``). ``c`` reads first byte (``00000010`` = 2). Output: ``2``.

Answer: 2

11. Question: In Python, what is the output of: ``from functools import reduce; print(reduce(lambda x, y: x + y, [1, 2, 3], 0))``?

Solution:

``reduce`` applies ``+`` with initial value ``0``: ``((0 + 1) + 2) + 3 = 6``. Output: ``6``.

Answer: 6

12. Question: In Java, what is the output of: ``try { throw new Exception("Error"); } catch (Exception e) { System.out.println("Caught"); } finally { System.out.println("Finally"); }``?

Solution:

Exception is caught, prints ``Caught``. ``finally`` always executes. Output: ``Caught Finally``.

Answer: Caught Finally

13. Question: In C++, what is the output of: ``vector<int> v = {1, 2, 3}; transform(v.begin(),`

`v.end(), v.begin(), [](int x) { return x * x; }); cout << v[1];`?`

Solution:

`transform` squares each element. `v = [1, 4, 9]`. `v[1] = 4`. Output: `4`.

Answer: 4

14. Question: In Python, what is the output of: ``x = [1, 2, 3]; y = iter(x); next(y); print(next(y))`?`

Solution:

`iter(x)` creates an iterator. First `next(y)` returns `1`. Second `next(y)` returns `2`. Output: `2`.

Answer: 2

15. Question: In C, what is the output of: ``int x = 5; int y = x++ + ++x; printf("%d %d", x, y);`?`

Solution:

Undefined behavior due to multiple modifications of `x` without sequence point. Possible: `x = 7`, `y = 12` (5 + 7). Output varies, assume common compiler: `7 12`.

Answer: 7 12 (compiler-dependent)

16. Question: In Java, what is the output of: ``List<Integer> list = Arrays.asList(1, 2, 3); list.replaceAll(x -> x + 1); System.out.println(list);`?`

Solution:

`replaceAll` increments each element. `list = [2, 3, 4]`. Output: `[2, 3, 4]`.

Answer: [2, 3, 4]

17. Question: In Python, what is the output of: ``import copy; x = [[1], [2]]; y = copy.deepcopy(x); y[0].append(3); print(x)`?`

Solution:

`deepcopy` creates independent copies. Modifying `y` doesn't affect `x`. Output: `[[1], [2]]`.

Answer: [[1], [2]]

18. Question: In C++, what is the output of: ``int x = 5; function<int(int)> f = [&](int a) { x += a; return x; }; cout << f(3);`?`

Solution:

Lambda captures `x` by reference. `f(3)` adds `3` to `x = 5`, returns `x = 8`. Output: `8`.

Answer: 8

19. Question: In C, what is the output of: ``struct S { int x; char c; } s = {10, 'A'}; printf("%d %c", s.x, s.c);`?`

Solution:

`s.x = 10`, `s.c = 'A'`. Output: `10 A`.

Answer: 10 A

20. Question: In Python, what is the output of: ``x = [1, 2, 3]; print(list(filter(lambda x: x % 2 == 1, x)))`?`

Solution:

`filter` selects odd numbers: `[1, 3]`. Output: `[1, 3]`.

Answer: [1, 3]

Data Structures (21?40)

21. Question: What is the time complexity of finding the lowest common ancestor (LCA) in a binary search tree with n nodes?

Solution:

Traverse from root to nodes, comparing values: $O(h)$, where h is height ($O(\log n)$ if balanced).

Answer: $O(\log n)$ (balanced)

22. Question: In Python, what is the output of this AVL tree insertion (simplified): ``class Node: def __init__(self, val): self.val = val; self.left = self.right = None; root = Node(10); root.left = Node(5); root.right = Node(15); root.right.right = Node(20); print(root.right.right.val)``?

Solution:

Tree structure: ``10 -> 15 -> 20``. ``root.right.right.val = 20``. Output: ``20``.

Answer: 20

23. Question: What is the time complexity of building a binary heap from an array of n elements?

Solution:

Heapify from bottom-up: $O(n)$.

Answer: $O(n)$

24. Question: In a graph with V vertices and E edges, what is the time complexity of detecting a cycle using DFS?

Solution:

DFS visits all vertices and edges: $O(V + E)$.

Answer: $O(V + E)$

25. Question: In Python, what is the output of: ``from collections import defaultdict; d = defaultdict(list); d[1].append(2); d[1].append(3); print(d)``?

Solution:

``defaultdict`` creates empty list for new keys. ``d[1] = [2, 3]``. Output: ``defaultdict(<class 'list'>, {1: [2, 3]})``.

Answer: `defaultdict(<class 'list'>, {1: [2, 3]})`

26. Question: What is the time complexity of finding the shortest path in an unweighted graph using BFS?

Solution:

BFS explores all vertices and edges: $O(V + E)$.

Answer: $O(V + E)$

27. Question: In a trie with n words of average length k , what is the time complexity to search for a word?

Solution:

Traverse characters: $O(k)$.

Answer: $O(k)$

28. Question: In Python, what is the output of: ``class Node: def __init__(self, val): self.val = val; self.next = None; head = Node(1); head.next = Node(2); head.next.next = Node(3); print(head.next.next.val)``?

Solution:

Linked list: ``1 -> 2 -> 3``. ``head.next.next.val = 3``. Output: ``3``.

Answer: 3

29. Question: What is the time complexity of Kruskal's algorithm for minimum spanning tree with V vertices and E edges?

Solution:

Sort edges $O(E \log E)$, union-find $O(E \cdot \alpha(V)) \approx O(E \log V)$. Total: $O(E \log V)$.

Answer: $O(E \log V)$

30. Question: In a segment tree for range sum queries with n elements, what is the time complexity of a query?

Solution:

Traverse height of tree: $O(\log n)$.

Answer: $O(\log n)$

31. Question: In Python, what is the output of: ``from heapq import heappush, heappop; h = []; heappush(h, 3); heappush(h, 1); heappush(h, 2); print(heappop(h))``?

Solution:

Min-heap: ``[1, 2, 3]``. ``heappop`` removes smallest: ``1``. Output: ``1``.

Answer: 1

32. Question: What is the time complexity of deleting a node from a red-black tree with n nodes?

Solution:

Rebalance after deletion: $O(\log n)$.

Answer: $O(\log n)$

33. Question: In a weighted graph, what is the time complexity of Bellman-Ford algorithm for single-source shortest paths?

Solution:

Relax edges $V-1$ times: $O(V * E)$.

Answer: $O(V * E)$

34. Question: In Python, what is the output of: ``from collections import deque; g = {1: [2, 3], 2: [4], 3: [4], 4: []}; q = deque([1]); visited = {1}; while q: n = q.popleft(); for nei in g[n]: if nei not in visited: visited.add(nei); q.append(nei); print(len(visited))``?

Solution:

BFS visits all nodes in graph: ``1, 2, 3, 4``. ``len(visited) = 4``. Output: ``4``.

Answer: 4

35. Question: What is the space complexity of storing a sparse graph with V vertices and E edges?

Solution:

Adjacency list: $O(V + E)$.

Answer: $O(V + E)$

36. Question: In a B-tree of order m with n keys, what is the time complexity of searching a key?

Solution:

Traverse height: $O(\log_m n)$.

Answer: $O(\log n)$

37. Question: In Python, what is the output of: `x = [(1, 'a'), (2, 'b'), (1, 'c')]; x.sort(); print(x)`?

Solution:

Sort by first element, then second: `[(1, 'a'), (1, 'c'), (2, 'b')]`. Output: `[(1, 'a'), (1, 'c'), (2, 'b')]`.

Answer: `[(1, 'a'), (1, 'c'), (2, 'b')]`

38. Question: What is the time complexity of finding strongly connected components using Kosaraju's algorithm?

Solution:

Two DFS passes and transpose: $O(V + E)$.

Answer: $O(V + E)$

39. Question: In a suffix tree for a string of length n , what is the time complexity to check if a substring exists?

Solution:

Traverse path matching substring: $O(m)$, where m is substring length.

Answer: $O(m)$

40. Question: In Python, what is the output of: `from collections import OrderedDict; d = OrderedDict([(1, 'a'), (2, 'b')]); d.move_to_end(1); print(list(d.keys()))`?

Solution:

`move_to_end(1)` moves key `1` to end. Keys: `[2, 1]`. Output: `[2, 1]`.

Answer: `[2, 1]`

Algorithms (41-60)

41. Question: What is the time complexity of the knapsack problem (0/1) using dynamic programming for n items and capacity W ?

Solution:

DP table of size $n \times W$: $O(n * W)$.

Answer: $O(n * W)$

42. Question: In Python, what is the output of this longest common subsequence: `def lcs(s1, s2): m, n = len(s1), len(s2); dp = [[0]*(n+1) for _ in range(m+1)]; for i in range(1, m+1): for j in range(1, n+1): if s1[i-1] == s2[j-1]: dp[i][j] = dp[i-1][j-1] + 1; else: dp[i][j] = max(dp[i-1][j], dp[i][j-1]); return dp[m][n]; print(lcs("ABC", "AC"))`?

Solution:

LCS of `"ABC"` and `"AC"` is `"AC"`. Length: `2`. Output: `2`.

Answer: 2

43. Question: What is the time complexity of matrix chain multiplication for n matrices using dynamic programming?

Solution:

DP table for all possible splits: $O(n^3)$.

Answer: $O(n^3)$

44. Question: In C++, what is the output of: `vector<int> v = {3, 1, 4, 1, 5}; int k = 2; multiset<int> s(v.begin(), v.begin()+k); int ans = *s.rbegin(); for(int i = k; i < v.size(); i++) { s.erase(s.find(v[i-k])); s.insert(v[i]); ans = max(ans, *s.rbegin()); } cout << ans;`?`

Solution:

Sliding window maximum using multiset. Max of `[3,1], [1,4], [4,1], [1,5]` is `4`. Output: `4`.

Answer: 4

45. Question: What is the time complexity of Floyd-Warshall algorithm for all-pairs shortest paths in a graph with V vertices?

Solution:

Three nested loops: $O(V^3)$.

Answer: $O(V^3)$

46. Question: In Python, what is the output of: `def kadane(arr): max_so_far = max_ending_here = arr[0]; for x in arr[1:]: max_ending_here = max(x, max_ending_here + x); max_so_far = max(max_so_far, max_ending_here); return max_so_far; print(kadane([-2, 1, -3, 4, -1, 2, 1]))`?`

Solution:

Kadane's algorithm finds max subarray sum: `[4, -1, 2, 1] = 6`. Output: `6`.

Answer: 6

47. Question: What is the time complexity of finding the median of two sorted arrays of size n each?

Solution:

Binary search to partition arrays: $O(\log n)$.

Answer: $O(\log n)$

48. Question: In C, what is the output of this KMP preprocessing: `void computeLPS(char *pat, int m, int *lps) { int len = 0, i = 1; lps[0] = 0; while(i < m) { if(pat[i] == pat[len]) { len++; lps[i++] = len; } else if(len > 0) len = lps[len-1]; else lps[i++] = 0; } } int lps[6]; computeLPS("AABAAC", 6, lps); printf("%d", lps[5]);`?`

Solution:

LPS for `"AABAAC"`: `[0, 1, 0, 1, 2, 0]`. `lps[5] = 0`. Output: `0`.

Answer: 0

49. Question: What is the time complexity of the A* search algorithm in a graph?

Solution:

Depends on heuristic, typically $O(b^d)$ where b is branching factor, d is depth.

Answer: $O(b^d)$

50. Question: In Python, what is the output of: ``def edit_distance(s1, s2): m, n = len(s1), len(s2); dp = [[0]*(n+1) for _ in range(m+1)]; for i in range(m+1): dp[i][0] = i; for j in range(n+1): dp[0][j] = j; for i in range(1, m+1): for j in range(1, n+1): if s1[i-1] == s2[j-1]: dp[i][j] = dp[i-1][j-1]; else: dp[i][j] = min(dp[i-1][j], dp[i][j-1], dp[i-1][j-1]) + 1; return dp[m][n]; print(edit_distance("cat", "cut"))``?

Solution:

Edit distance from ``"cat"`` to ``"cut"`` is 1 (replace ``a`` with ``u``). Output: ``1``.

Answer: 1

51. Question: What is the time complexity of finding the longest palindromic substring using dynamic programming?

Solution:

DP table for all substrings: $O(n^2)$.

Answer: $O(n^2)$

52. Question: In C++, what is the output of: ``int n = 5; vector<vector<int>> dp(n+1, vector<int>(n+1)); for(int i = 0; i <= n; i++) dp[i][0] = dp[0][i] = 1; for(int i = 1; i <= n; i++) for(int j = 1; j <= n; j++) dp[i][j] = dp[i-1][j] + dp[i][j-1]; cout << dp[n][n];``?

Solution:

DP computes number of paths in $n \times n$ grid: ``dp[5][5] = 252`` (binomial coefficient). Output: ``252``.

Answer: 252

53. Question: What is the space complexity of the Rabin-Karp string matching algorithm?

Solution:

Rolling hash and pattern storage: $O(1)$ extra space.

Answer: $O(1)$

54. Question: In Python, what is the output of: ``def max_profit(prices): n = len(prices); if n < 2: return 0; dp = [0]*n; min_price = prices[0]; for i in range(1, n): min_price = min(min_price, prices[i]); dp[i] = max(dp[i-1], prices[i] - min_price); return dp[n-1]; print(max_profit([7, 1, 5, 3, 6, 4]))``?

Solution:

Max profit by buying at ``1``, selling at ``6``: ``6 - 1 = 5``. Output: ``5``.

Answer: 5

55. Question: What is the time complexity of computing the convex hull of n points using Graham's scan?

Solution:

Sort points $O(n \log n)$, scan $O(n)$. Total: $O(n \log n)$.

Answer: $O(n \log n)$

56. Question: In C, what is the output of: ``int n = 4; int dp[5]; dp[0] = 1; for(int i = 1; i <= n; i++)`

{ dp[i] = 0; for(int j = 0; j < i; j++) dp[i] += dp[j] * dp[i-j-1]; } printf("%d", dp[n]);`?

Solution:

Catalan number for `n = 4`: `dp[4] = 14` (number of BSTs). Output: `14`.

Answer: 14

57. Question: What is the time complexity of the Hungarian algorithm for bipartite matching with n vertices?

Solution:

Maximum flow approach: $O(n^3)$.

Answer: $O(n^3)$

58. Question: In Python, what is the output of: ``def count_inversions(arr): def merge_and_count(l, r): inv = 0; res = []; i = j = 0; while i < len(l) and j < len(r): if l[i] <= r[j]: res.append(l[i]); i += 1; else: res.append(r[j]); inv += len(l)-i; j += 1; res.extend(l[i:]); res.extend(r[j:]); return res, inv; def sort_and_count(arr): if len(arr) <= 1: return arr, 0; mid = len(arr)//2; l, inv_l = sort_and_count(arr[:mid]); r, inv_r = sort_and_count(arr[mid:]); merged, inv_m = merge_and_count(l, r); return merged, inv_l + inv_r + inv_m; _, inv = sort_and_count([3, 1, 2]); print(inv)``?

Solution:

Inversions in `[3, 1, 2]`: `(3, 1), (3, 2)`. Count: `2`. Output: `2`.

Answer: 2

59. Question: What is the time complexity of finding the k-th smallest element in two sorted arrays of sizes m and n?

Solution:

Binary search on partitions: $O(\log(\min(m, n)))$.

Answer: $O(\log(\min(m, n)))$

60. Question: In C++, what is the output of: ``int n = 3; vector<int> dp(n+1); dp[0] = 1; for(int i = 1; i <= n; i++) for(int j = 1; j <= i; j++) dp[i] += dp[i-j]; cout << dp[n];``?

Solution:

Number of ways to sum to `n` with positive integers: `dp[3] = 4` (1+1+1, 1+2, 2+1, 3). Output: `4`.

Answer: 4

Database Basics (61?70)

61. Question: What is the result of this SQL query: ``SELECT d.department_name, AVG(e.salary) FROM employees e JOIN departments d ON e.department_id = d.id GROUP BY d.department_name HAVING COUNT(e.id) > 5 ORDER BY AVG(e.salary) DESC;``?

Solution:

Joins `employees` and `departments`, groups by `department\_name`, computes average `salary`, filters departments with >5 employees, sorts by average salary descending.

Answer: Department names and average salaries for departments with >5 employees, sorted

high to low

62. Question: What does ``SELECT c.customer_name, COUNT(o.order_id) FROM customers c LEFT JOIN orders o ON c.id = o.customer_id GROUP BY c.customer_name;`` do?

Solution:

Counts orders per customer, including customers with zero orders.

Answer: Customer names and their order counts

63. Question: What is the output of: ``SELECT product_name, price FROM products WHERE price > (SELECT AVG(price) FROM products WHERE category = products.category);``?

Solution:

Retrieves ``product_name`` and ``price`` for products priced above their category's average.

Answer: Products above category average price

64. Question: What does ``CREATE TRIGGER update_stock AFTER INSERT ON orders FOR EACH ROW UPDATE products SET stock = stock - NEW.quantity WHERE id = NEW.product_id;`` do?

Solution:

Automatically reduces ``stock`` in ``products`` by ``quantity`` after an order is inserted.

Answer: Updates product stock on order insertion

65. Question: What is the result of: ``SELECT e1.name FROM employees e1 WHERE e1.salary > (SELECT AVG(e2.salary) FROM employees e2 WHERE e2.manager_id = e1.id);``?

Solution:

Retrieves names of employees with salaries above the average salary of their direct reports.

Answer: Employees earning above their team's average

66. Question: What does ``EXPLAIN SELECT * FROM orders WHERE customer_id = 100;`` do?

Solution:

Provides query execution plan, showing how the database processes the query.

Answer: Shows query plan

67. Question: What is the output of: ``SELECT category, MAX(price) FROM products GROUP BY category UNION SELECT 'Total', MAX(price) FROM products;``?

Solution:

Lists maximum ``price`` per ``category``, plus overall maximum ``price`` labeled as ``Total``.

Answer: Category max prices and overall max price

68. Question: What does ``ALTER TABLE employees ADD CONSTRAINT unique_email UNIQUE (email);`` do?

Solution:

Adds a unique constraint to ``email`` column, preventing duplicate emails.

Answer: Enforces unique emails

69. Question: What is the result of: ``SELECT name FROM students s WHERE EXISTS (SELECT`

1 FROM courses c WHERE c.student_id = s.id AND c.grade = 'A');`?

Solution:

Retrieves `name` of students who have at least one course with grade `A`.

Answer: Students with A-grade courses

70. Question: What does `CREATE VIEW high\_salary AS SELECT name, salary FROM employees WHERE salary > 100000;` do?

Solution:

Creates a virtual table showing `name` and `salary` for employees with `salary > 100000`.

Answer: Creates high-salary view

Computer Networks (71?80)

71. Question: What is the purpose of the Transport layer in the OSI model?

Solution:

Provides reliable data transfer, flow control, and error correction.

Answer: Reliable data transfer

72. Question: What is the role of IPSec in network security?

Solution:

Provides encryption and authentication for IP packets.

Answer: Secures IP communication

73. Question: What is the difference between TCP and SCTP?

Solution:

TCP: single stream, reliable; SCTP: multi-stream, supports multi-homing.

Answer: TCP: single stream, SCTP: multi-stream

74. Question: What is the purpose of the DHCP protocol?

Solution:

Dynamically assigns IP addresses to devices on a network.

Answer: IP address assignment

75. Question: What is the role of the TTL field in an IP header?

Solution:

Limits packet lifetime to prevent infinite loops (decremented per hop).

Answer: Prevents routing loops

76. Question: What is the difference between stateful and stateless firewalls?

Solution:

Stateful: tracks connection states; Stateless: filters packets based on rules.

Answer: Stateful: connection-aware, Stateless: rule-based

77. Question: What is the purpose of the RST flag in TCP?

Solution:

Resets a connection due to errors or invalid segments.

Answer: Connection reset

78. Question: What is the role of SNMP in network management?

Solution:

Simple Network Management Protocol monitors and manages network devices.

Answer: Network monitoring

79. Question: What is the maximum throughput of a network with 100 Mbps bandwidth and 10 ms RTT, assuming TCP with window size 64 KB?

Solution:

Throughput = Window Size / RTT = $(64 * 1024 * 8) / (10 / 1000) = 51.2 \text{ Mbps}$.

Answer: 51.2 Mbps

80. Question: What is the purpose of VLAN tagging in Ethernet networks?

Solution:

Adds identifiers to frames to separate traffic on virtual LANs.

Answer: Separates VLAN traffic

Operating Systems (81-90)

81. Question: What is the difference between spinlock and semaphore in an OS?

Solution:

Spinlock: busy-waits in a loop; Semaphore: blocks threads, supports counting.

Answer: Spinlock: busy-wait, Semaphore: blocking

82. Question: What is the role of the TLB (Translation Lookaside Buffer) in virtual memory?

Solution:

Caches recent virtual-to-physical address translations to speed up access.

Answer: Speeds address translation

83. Question: What is the Banker's algorithm used for in an OS?

Solution:

Prevents deadlock by ensuring safe resource allocation.

Answer: Deadlock avoidance

84. Question: What is the difference between user-level and kernel-level threads?

Solution:

User-level: managed by application, no kernel involvement; Kernel-level: OS-managed, supports parallelism.

Answer: User-level: app-managed, Kernel-level: OS-managed

85. Question: What is the purpose of copy-on-write in memory management?

Solution:

Delays copying shared memory pages until a write occurs, saving resources.

Answer: Optimizes memory usage

86. Question: What is the role of the inverted page table in an OS?

Solution:

Maps physical pages to virtual pages, reducing memory overhead.

Answer: Efficient page mapping

87. Question: What is the difference between hard and soft real-time systems?

Solution:

Hard: strict deadlines, failure is critical; Soft: deadlines important but not critical.

Answer: Hard: critical, Soft: non-critical

88. Question: What is the purpose of the monitor in process synchronization?

Solution:

Ensures mutual exclusion and provides condition variables for synchronization.

Answer: Synchronization construct

89. Question: What is the role of the I/O scheduler in an OS?

Solution:

Optimizes disk access by reordering I/O requests.

Answer: Optimizes disk I/O

90. Question: What is the difference between demand paging and prepaging?

Solution:

Demand paging: loads pages on demand; Prepaging: loads pages in advance to reduce faults.

Answer: Demand: on-demand, Prepaging: anticipatory

AI/ML Foundations (91?100)

91. Question: What is the difference between L1 and L2 regularization in ML?

Solution:

L1: adds absolute weights (sparsity); L2: adds squared weights (smoothness).

Answer: L1: sparsity, L2: smoothness

92. Question: What is the purpose of early stopping in neural network training?

Solution:

Halts training when validation performance stops improving to prevent overfitting.

Answer: Prevents overfitting

93. Question: What is the role of batch normalization in deep learning?

Solution:

Normalizes layer inputs to stabilize and accelerate training.

Answer: Stabilizes training

94. Question: What is the difference between precision and recall in ML evaluation?

Solution:

Precision: $TP/(TP+FP)$ (accuracy of positive predictions); Recall: $TP/(TP+FN)$ (coverage of actual positives).

Answer: Precision: accuracy, Recall: coverage

95. Question: What is the purpose of dropout in neural networks?

Solution:

Randomly deactivates neurons during training to prevent overfitting.

Answer: Reduces overfitting

96. Question: What is the difference between CNN and RNN architectures?

Solution:

CNN: processes spatial data (e.g., images); RNN: processes sequential data (e.g., time series).

Answer: CNN: spatial, RNN: sequential

97. Question: What is the role of the attention mechanism in transformers?

Solution:

Weights input elements based on relevance, improving context understanding.

Answer: Focuses on relevant inputs

98. Question: What is the purpose of the F1 score in ML evaluation?

Solution:

Harmonic mean of precision and recall, balances both metrics.

Answer: Balances precision and recall

99. Question: What is the difference between generative and discriminative models in ML?

Solution:

Generative: models joint probability $P(x,y)$; Discriminative: models conditional $P(y|x)$.

Answer: Generative: joint, Discriminative: conditional

100. Question: What is the purpose of transfer learning in deep learning?

Solution:

Uses pre-trained models to improve performance on new tasks with less data.

Answer: Leverages pre-trained models